



Software Requirements Specification (SRS)

PayFlow Malawi - Integrated Payroll & Deduction Management Platform

Multi-Tenant Automated Deduction and HR Management System

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for PayFlow Malawi, a comprehensive multi-tenant platform that integrates payroll management, HR modules, automated deduction processing, and mobile customer services for employers, lending institutions, and employees in Malawi.

1.2 Document Conventions

- **Shall/Must:** Mandatory requirements
- **Should:** Recommended requirements
- **May:** Optional requirements
- **Institution:** Organization using the platform (employers, SACCOs, financial institutions)
- **Super Admin:** Core platform administrator
- **Institution Admin:** Administrator for specific institution

1.3 Intended Audience

- System developers and architects
- Project managers
- Quality assurance teams
- Business stakeholders
- System administrators
- Institutional clients

1.4 Product Scope

PayFlow Malawi is a comprehensive multi-tenant platform that provides:

- HR Management modules for employers
- Automated payroll deduction processing
- SACCO and financial institution deduction management
- Employee self-service mobile applications
- Integration capabilities with existing payroll/HR and lending systems
- Affordability assessment and deduction processing workflows



2. Overall Description

2.1 Product Perspective

PayFlow Malawi operates as a centralized multi-tenant platform serving:

- **Employers:** Government ministries, private companies, NGOs
- **SACCOs/Financial Institutions:** Savings cooperatives, microfinance institutions
- **Employees:** Civil servants, private sector workers
- **Super Administrators:** Core platform management team

2.2 Product Functions

- Multi-tenant architecture with institution-based data segregation
- Comprehensive HR management modules
- Payroll processing and integration
- Automated deduction management for shares and repayments
- Affordability assessment engine
- SACCO/Financial institution integration portals
- Employee mobile self-service with biometric security
- Real-time reporting and analytics
- API integration capabilities with external lending systems

2.3 User Classes and Characteristics

2.3.1 Super Admin Level

- **Platform Super Admin:** Complete system oversight, institution setup, system maintenance

2.3.2 Institution Level - Employers

- **Employer Admin:** Institution management, user creation, system configuration
- **HR Manager:** Employee management, payroll processing, reporting
- **Payroll Officer:** Payroll processing, deduction management
- **HR Officer:** Employee data management, basic reporting

2.3.3 Institution Level - SACCOs/Financial Institutions

- **Institution Admin:** User management, system configuration, reporting
- **Operations Officer:** Deduction monitoring, affordability checks, member management
- **Operations Manager:** Deduction reconciliation, analytics, reporting
- **Customer Service Rep:** Member inquiries, account management



2.3.4 End Users

- **Employees:** Mobile app users, self-service capabilities

2.4 Operating Environment

- **Platform:** Multi-tenant architecture (self-hosted or cloud)
 - **Server:** Self-hosted infrastructure or cloud-based (AWS/Azure/Google Cloud)
 - **Database:** PostgreSQL/MySQL with institution-based data segregation
 - **Web Application:** Responsive web interface
 - **Mobile App:** Android/iOS native applications with biometric authentication
 - **API:** RESTful API services for external system integration
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3. System Architecture

3.1 Multi-Tenant Architecture

3.1.1 Institution-Based Data Segregation

3.1.1.1 The system shall implement row-level security using institution identifiers for data isolation.

3.1.1.2 The system shall ensure all database records contain institution id for proper data segregation.

3.1.1.3 The system shall provide application-level access control based on user's institution association.

3.1.2 Institution Types

3.1.2.1 The system shall support three primary institution types:

- **Employer Institutions:** Organizations managing employees and payroll
- **SACCO/Financial Institutions:** Organizations processing shares and deductions
- **Hybrid Institutions:** Organizations serving both as employers and financial institutions

3.2 System Components

3.2.1 Core Platform Components

- **Super Admin Console:** Platform management and institution provisioning
- **Institution Management Service:** Institution lifecycle management
- **Authentication & Authorization Service:** Multi-institution security
- **Integration Hub:** API management and external system connectivity



- **Notification Service:** SMS/Email communication
- **Reporting Engine:** Cross-institution analytics and reporting

3.2.2 Institution-Specific Components

- **HR Management Module:** Employee lifecycle management
 - **Payroll Processing Module:** Salary calculation and processing
 - **Deduction Management Module:** Automated deduction processing
 - **SACCO Portal:** Share and deduction management
 - **Mobile API Gateway:** Employee self-service backend
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4. System Features

4.1 Super Admin Management

4.1.1 Description

The system shall provide comprehensive platform management capabilities for super administrators.

4.1.2 Functional Requirements

4.1.2.1 The system shall allow super admins to create and manage institution accounts:

- Employer institution creation
- SACCO/Financial institution creation
- Hybrid institution configuration
- Institution status management (Active, Suspended, Terminated)

4.1.2.2 The system shall provide institution monitoring capabilities:

- Usage analytics per institution
- Performance monitoring
- Resource utilization tracking
- System health monitoring

4.1.2.3 The system shall support institution configuration management:

- Feature enablement/disablement per institution
- Integration endpoint configuration
- Compliance settings management
- Data retention policies

4.2 Institution Management - Employers



4.2.1 Description

The system shall provide comprehensive institution management for employer organizations.

4.2.2 Functional Requirements

4.2.2.1 The system shall allow employer admins to manage institution settings:

- Company profile management
- Department and location setup
- Payroll configuration
- Deduction policies and rules
- Integration settings

4.2.2.2 The system shall provide user management within institution:

- User role assignment (HR Manager, Payroll Officer, HR Officer)
- User permissions configuration
- User activity monitoring
- Password policy enforcement

4.3 HR Management Module

4.3.1 Description

The system shall provide comprehensive HR management capabilities for employers without existing HR systems.

4.3.2 Functional Requirements

4.3.2.1 Employee Information Management:

- Complete employee profile management
- Employee onboarding workflows
- Document management (contracts, certificates, IDs)
- Employee hierarchy management
- Employee number assignment and management

4.3.2.2 Organizational Structure Management:

- Department creation and management
- Position and job title management
- Reporting structure configuration
- Location and branch management

4.3.2.3 Leave Management:



- Leave policy configuration
- Leave application workflows
- Leave balance tracking
- Leave calendar management

4.3.2.4 Attendance Management:

- Time tracking integration
- Attendance reporting
- Overtime calculation
- Shift management

4.4 Payroll Processing Module

4.4.1 Description

The system shall provide comprehensive payroll processing capabilities with deduction integration.

4.4.2 Functional Requirements

4.4.2.1 Payroll Configuration:

- Salary structure setup
- Tax calculation configuration (PAYE)
- Statutory deduction setup (pension contributions)
- Allowance and benefit configuration

4.4.2.2 Payroll Processing:

- Automated payroll calculation
- Batch processing capabilities
- Payroll approval workflows
- Payroll reversal and correction capabilities

4.4.2.3 Deduction Integration:

- Automatic deduction calculation from SACCO/financial institutions
- Manual deduction entry capabilities
- Deduction approval workflows
- Deduction reconciliation with institutions

4.4.2.4 Payroll File Generation:

- Bank transfer file generation
- Payroll register reports
- Tax authority submissions



- Statutory reporting

4.5 Integration Management

4.5.1 Description

The system shall provide comprehensive integration capabilities for existing payroll/HR and lending systems.

4.5.2 Functional Requirements

4.5.2.1 API Integration:

- RESTful API for employee data synchronization
- Payroll data import/export capabilities
- Real-time webhook notifications for deduction updates
- Integration with external lending systems for deduction requests

4.5.2.2 File-Based Integration:

- Excel/CSV file upload capabilities
- Automated file processing with validation
- Error handling and exception reporting
- File format templating and validation

4.5.2.3 Manual Upload Features:

- Deduction upload by employers
- Payroll data verification and approval
- Exception handling workflows
- Comprehensive audit trail maintenance

4.5.2.4 External Lending System Integration:

- API interfaces for loan origination systems
- Deduction request processing from external systems
- Real-time affordability checks for external requests
- Status updates and notifications to external systems

4.6 SACCO/Financial Institution Portal

4.6.1 Description

The system shall provide dedicated portals for SACCOs and financial institutions to manage deduction processes.

4.6.2 Functional Requirements

4.6.2.1 Institution Management:



- Institution profile management
- Share product configuration
- Fee structure setup
- Member category management

4.6.2.2 Deduction Management:

- Share deduction request creation
- Repayment deduction processing
- Employer communication tools
- Deduction status tracking and monitoring

4.6.2.3 Member Management:

- Member profile management within employer organizations
- Share balance tracking
- Deduction history monitoring
- Member communication tools

4.6.2.4 Integration Management:

- External lending system integration setup
- API endpoint configuration for loan systems
- Webhook management for status updates
- Data synchronization with core banking systems

4.7 Affordability Assessment Engine

4.7.1 Description

The system shall provide comprehensive affordability assessment capabilities for institutions and external systems.

4.7.2 Functional Requirements

4.7.2.1 Affordability Calculation:

- Real-time salary verification across employers
- Existing deduction analysis across all institutions
- Regulatory compliance checking (maximum deduction limits)
- Risk assessment scoring based on payment history

4.7.2.2 Multi-Source Assessment:

- Cross-institution deduction checking
- Historical payment behavior analysis
- Employment stability verification



- Income consistency tracking

4.7.2.3 Assessment Reporting:

- Detailed affordability reports with recommendations
- Risk score explanations and factors
- Compliance documentation for regulatory requirements
- Real-time API responses for external system queries

4.7.2.4 External System Integration:

- API endpoints for external affordability checks
- Real-time assessment for loan origination systems
- Batch processing for bulk assessments
- Historical data provision for credit decisions

4.8 Employee Mobile Application

4.8.1 Description

The system shall provide a comprehensive mobile application for employee self-service with enhanced security features.

4.8.2 Functional Requirements

4.8.2.1 Authentication and Security:

- Initial login using employee number or phone number
- SMS OTP verification for account setup and critical actions
- Biometric authentication (fingerprint, facial recognition) stored locally
- App-level security with biometric protection
- Automatic session timeout and re-authentication

4.8.2.2 Account Management:

- Profile viewing and limited updates
- Contact information management
- Emergency contact updates
- Document viewing capabilities

4.8.2.3 Deduction Monitoring:

- Real-time deduction tracking across all institutions
- Payment history viewing with detailed breakdowns
- Outstanding balance monitoring
- Deduction schedule viewing and projections

4.8.2.4 Affordability Services:



- Self-service affordability assessment
- Pre-qualification checks for new deductions
- Comparison of available financial products
- Historical affordability trend analysis

4.8.2.5 Financial Information:

- Salary information and payslips
- Tax deduction summaries
- Pension contribution tracking
- Total deduction summaries across all institutions

4.8.2.6 Support Services:

- In-app customer support system (ticketing)
- FAQ and help documentation
- Contact information for all associated institutions
- Support ticket tracking and status updates

4.9 Reporting and Analytics

4.9.1 Description

The system shall provide comprehensive reporting capabilities for all user types.

4.9.2 Functional Requirements

4.9.2.1 Employer Reports:

- Payroll processing summary reports
- Deduction reconciliation reports by institution
- Employee deduction analysis
- Compliance and statutory reporting

4.9.2.2 SACCO/Financial Institution Reports:

- Member deduction portfolio analytics
- Collection efficiency tracking
- Member behavior analysis
- Affordability assessment summaries

4.9.2.3 Super Admin Reports:

- Platform usage analytics across institutions
- System performance metrics
- Institution activity monitoring
- Revenue and billing summaries



4.9.2.4 Employee Reports:

- Personal deduction history across all institutions
- Affordability assessment history
- Payment schedules and projections
- Financial wellness summaries

4.10 Notification and Communication

4.10.1 Description

The system shall provide comprehensive notification capabilities through multiple channels.

4.10.2 Functional Requirements

4.10.2.1 SMS Notifications:

- OTP delivery for authentication
- Deduction processing confirmations
- Payment due reminders
- Critical account alerts

4.10.2.2 Email Notifications:

- Monthly deduction summaries
- System maintenance notifications
- Account status updates
- Report delivery notifications

4.10.2.3 Push Notifications (Mobile App):

- Real-time deduction updates
- Affordability check results
- Payment confirmations
- Security alerts

4.10.2.4 In-App Communication:

- Support ticket system
 - Institution announcements
 - System notifications
 - Help and guidance messages
-



5. External Interface Requirements

5.1 User Interfaces

5.1.1 Web Application Interface

5.1.1.1 The system shall provide institution-specific web interfaces with:

- Consistent PayFlow Malawi branding
- Role-based dashboard customization
- Responsive design for all devices
- Accessibility compliance (WCAG 2.1)

5.1.1.2 The system shall provide consistent navigation across all institution interfaces:

- Institution-specific menu structures based on type
- Breadcrumb navigation
- Global search functionality
- Quick action buttons and shortcuts

5.1.2 Mobile Application Interface

5.1.2.1 The system shall provide native mobile applications for:

- Android devices (API level 21+)
- iOS devices (iOS 12+)
- Cross-platform functionality consistency

5.1.2.2 The system shall provide mobile-optimized features:

- Biometric authentication support (stored locally)
- Offline data caching for critical information
- SMS OTP integration
- Camera integration for document scanning

5.2 API Interfaces

5.2.1 RESTful API Services

5.2.1.1 The system shall provide comprehensive API endpoints for:

- Employee data management and synchronization
- Payroll integration and processing
- Deduction processing and status updates
- Affordability assessment requests

5.2.1.2 The system shall implement API security with:

- OAuth 2.0 authentication



- Institution-based access control
- Rate limiting per institution
- Request/response encryption

5.2.2 External System Integration APIs

5.2.2.1 The system shall provide APIs for external lending systems:

- Affordability check endpoints
- Deduction request submission
- Status notification webhooks
- Member verification services

5.2.2.2 The system shall provide webhook capabilities for:

- Real-time deduction status updates
- Payment confirmation notifications
- Affordability assessment results
- System error notifications

5.3 Third-Party Integrations

5.3.1 Communication Services

5.3.1.1 The system shall integrate with:

- SMS gateway services for OTP delivery
- Email service providers
- Push notification services
- Communication logging services

5.3.2 External System Integration

5.3.2.1 The system shall provide integration capabilities with:

- Core banking systems
- Loan origination systems
- Credit bureau services
- Government payroll systems

6. Non-Functional Requirements

6.1 Performance Requirements

6.1.1 Scalability



6.1.1.1 The system shall support:

- Up to 500 concurrent users per institution
- 50,000+ employees across all institutions
- 200+ active institutions simultaneously
- 10M+ transactions per month

6.1.2 Response Time

6.1.2.1 The system shall provide:

- Web interface response time < 2 seconds
- Mobile app response time < 3 seconds
- API response time < 1 second
- SMS OTP delivery within 30 seconds

6.2 Security Requirements

6.2.1 Data Security

6.2.1.1 The system shall implement:

- End-to-end encryption for all sensitive data
- Database encryption at rest
- Secure communication protocols (TLS 1.3)
- Regular security audits and vulnerability assessments

6.2.2 Multi-Institution Security

6.2.2.1 The system shall ensure:

- Complete institution data isolation using row-level security
- Cross-institution access prevention
- Institution-specific security policies
- Secure API access with proper authentication

6.2.3 Mobile Security

6.2.3.1 The system shall implement:

- Local biometric data storage (not transmitted)
- App-level security with biometric protection
- Certificate pinning for API communications
- Secure local data storage with encryption

6.3 Reliability Requirements

6.3.1 Availability



6.3.1.1 The system shall maintain:

- 99.5% uptime SLA
- 24/7 system availability for critical functions
- Planned maintenance windows < 4 hours monthly
- Disaster recovery capabilities

6.3.2 Data Integrity

6.3.2.1 The system shall ensure:

- Zero data loss tolerance for financial transactions
- Automated backup procedures
- Point-in-time recovery capabilities
- Transaction logging and audit trails

6.4 Usability Requirements

6.4.1 User Experience

6.4.1.1 The system shall provide:

- Intuitive user interfaces across all platforms
- Consistent user experience across all modules
- Comprehensive help documentation
- Multi-language support (English, Chichewa)

6.4.2 Mobile Usability

6.4.2.1 The system shall support:

- Simple and secure authentication flow
- Offline access to critical information
- Intuitive navigation and information hierarchy
- Accessibility features for diverse user capabilities

7. Other Requirements

7.1 Deployment Requirements

7.1.1 Self-Hosting Support

7.1.1.1 The system shall support self-hosted deployment with:

- Comprehensive installation documentation
- System requirements specification



- Database setup and configuration scripts
- Security hardening guidelines

7.1.2 Cloud Deployment

7.1.2.1 The system shall support cloud deployment on:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform
- Private cloud infrastructures

7.2 Regulatory Requirements

7.2.1 Financial Regulations

7.2.1.1 The system shall comply with:

- Reserve Bank of Malawi regulations
- SACCO regulatory requirements
- Anti-money laundering requirements
- Financial data protection standards

7.2.2 Employment Regulations

7.2.2.1 The system shall comply with:

- Employment Act of Malawi
- Labour Relations Act
- Pension and retirement regulations
- Tax and statutory deduction requirements

7.3 Business Requirements

7.3.1 Multi-Institution Business Model

7.3.1.1 The system shall support:

- Subscription-based pricing models
- Usage-based billing options
- Institution-specific feature enablement
- Service level agreement management

7.3.2 Integration Business Model

7.3.2.1 The system shall enable:

- Rapid institution onboarding
- API-based external system integration



- White-label integration capabilities
- Partner ecosystem development

7.4 Technical Requirements

7.4.1 Development Standards

7.4.1.1 The system shall follow:

- Microservices architecture principles
- API-first development approach
- Automated testing frameworks
- Continuous integration/deployment practices

7.4.2 Monitoring and Logging

7.4.2.1 The system shall implement:

- Comprehensive application logging
 - Real-time monitoring and alerting
 - Performance metrics tracking
 - Security event logging and analysis
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Appendices

Appendix A: Glossary

- **PayFlow Malawi:** The comprehensive multi-institution platform
- **Institution:** Organization using the platform (employer, SACCO, financial institution)
- **Super Admin:** Core platform administrator
- **Institution Admin:** Institution-level administrator
- **SACCO:** Savings and Credit Cooperative Organization
- **OTP:** One-Time Password
- **API:** Application Programming Interface

Appendix B: Institution Onboarding Process

1. Super Admin creates institution account
2. Institution receives onboarding credentials
3. Institution completes profile setup
4. System configuration and user setup
5. Employee data import/setup
6. Integration setup and testing



7. Go-live and support transition

Appendix C: Mobile App Security Flow

1. User downloads and installs mobile app
2. Initial registration using employee number/phone
3. SMS OTP verification for account setup
4. Biometric enrollment (stored locally on device)
5. Subsequent logins use biometric authentication
6. OTP required only for critical actions or security events
- 7.

Appendix D: Integration Scenarios

1. **Greenfield Implementation:** Full PayFlow Malawi deployment
 2. **Payroll Integration:** Integration with existing payroll systems
 3. **Lending System Integration:** API integration with loan origination systems
 4. **Hybrid Deployment:** Partial system integration with existing infrastructure
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